

Post-Doc Interview Presentation

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Coordinate Descent for SLOPE (Latest Work)

Previous Work

Ongoing Work

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Coordinate Descent for SLOPE

The Problem

SLOPE is a sparsity-inducing model with appealing properties, but the best algorithms (up til now) for solving SLOPE are slow.

Our Contribution

A hybrid algorithm based on coordinate descent (CD) and proximal gradient descent.

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Our Contribution

A hybrid algorithm based on coordinate descent (CD) and proximal gradient descent.

A collaboration with Quentin Klopfenstein, Mathurin Massias, and Jonas Wallin.



Sorted L-One Penalized Estimation (SLOPE)

For a design matrix $X \in \mathbb{R}^{n \times p}$ and response vector $y \in \mathbb{R}^n$, the solution to SLOPE is

$$\beta^* \in \arg \min_{\beta \in \mathbb{R}^p} \left\{ P(\beta) = \frac{1}{2} \|y - X\beta\|^2 + J(\beta) \right\}$$

where

$$J(\beta) = \sum_{j=1}^p \lambda_j |\beta_{(j)}|$$

is the sorted ℓ_1 norm, defined through

$$|\beta_{(1)}| \geq |\beta_{(2)}| \geq \cdots \geq |\beta_{(p)}|, \quad (1)$$

with λ being a fixed non-increasing and non-negative sequence.

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Special Cases

- $\lambda_1 = \cdots = \lambda_p \rightarrow \ell_1$ (the lasso penalty)
- $\lambda_1 > \lambda_2 = \cdots = \lambda_p = 0 \rightarrow \ell_\infty$

Properties

SLOPE has many appealing properties:

- **Clustering** (Bogdan, Dupuis, et al. 2022; Schneider and Tardivel 2020; Figueiredo and Nowak 2016)
- **Control of false discovery rate** (Bogdan, Berg, Su, et al. 2013; Bogdan, Berg, Sabatti, et al. 2015)
- **Recovery of sparsity and ordering patterns** (Bogdan, Dupuis, et al. 2022)
- **Convexity**

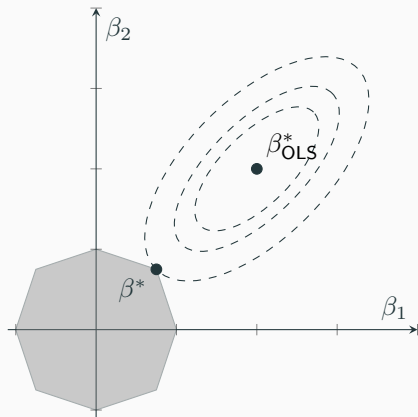


Figure 1: The SLOPE solution seen as a constrained problem.

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So why isn't SLOPE more popular?

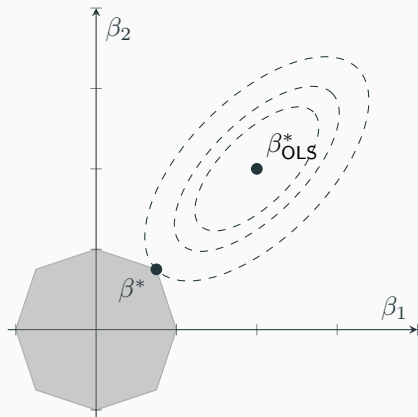


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Coordinate Descent

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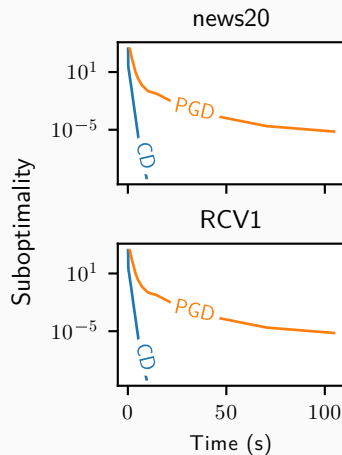


Figure 2: Coordinate descent versus proximal gradient descent for the lasso.

Coordinate Descent

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- Simple optimization method: at each iteration, update a single coordinate (coefficient).

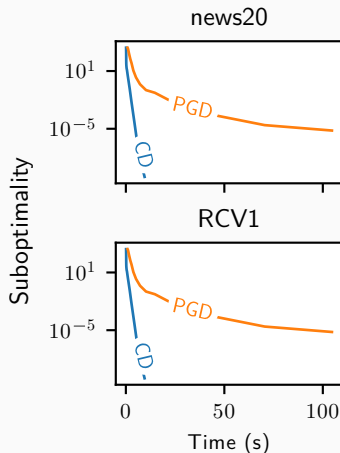


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Coordinate Descent and Inseparability

- Unfortunately, we cannot use basic coordinate descent for SLOPE since the sorted ℓ_1 norm is **inseparable**:

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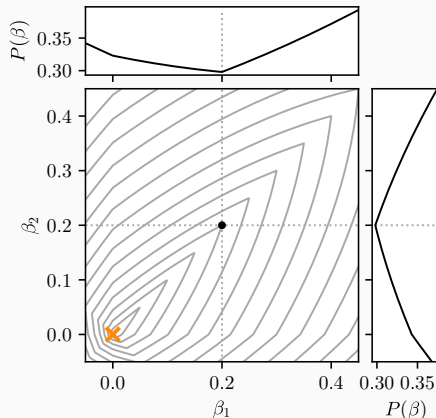


Figure 3: A *naive* coordinate descent algorithm cannot advance from the current iterate (●) to reach the optimum (✕).

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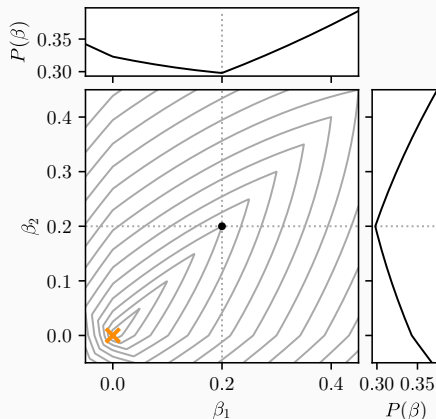


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- But if we fix the clusters, we have separability and can solve SLOPE using coordinate descent.
- Idea:** Alternate between gradient descent steps (identify the clusters) and coordinate descent steps **on the clusters** (converge quickly).

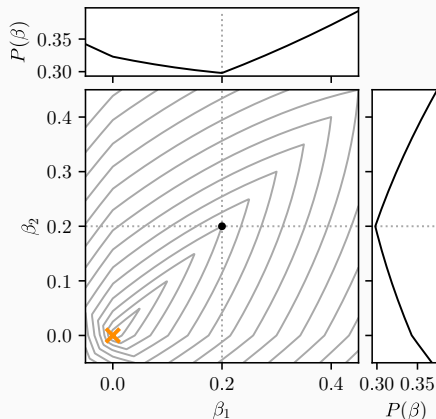


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Hybrid Algorithm

- Every v th iteration, take a full proximal gradient step. This allows clusters to split (or merge).
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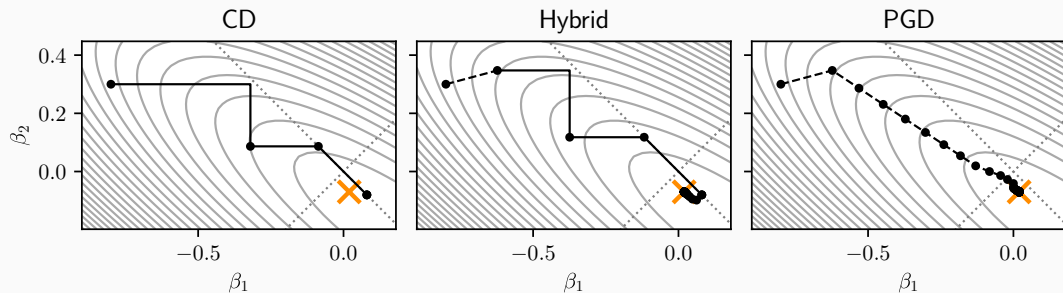


Figure 4: Our algorithm (hybrid) is a combination of CD and PGD.

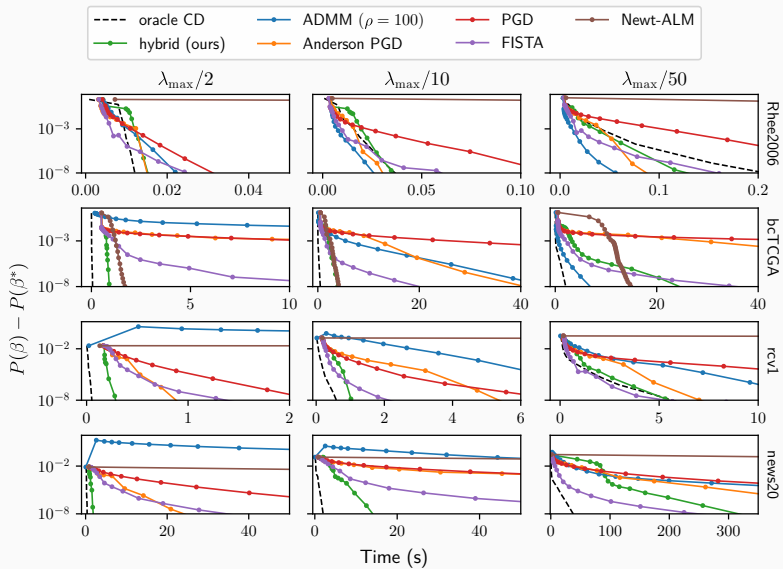


Figure 5: Benchmarks on real data

Previous Work

The Strong Screening Rule for SLOPE

Basic idea:

- When $p \gg n$, SLOPE and lasso solutions have small support.
- If we can estimate the support (before fitting the model), we save a lot of time.
- If the screening method is cheap, we have a net gain.

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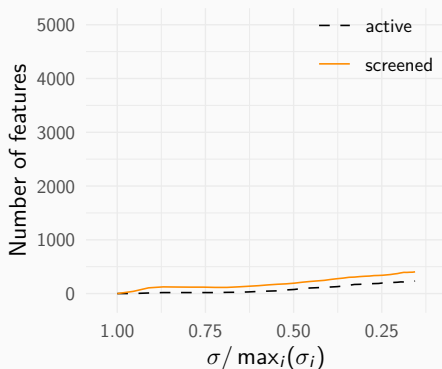


Figure 6: Number of features screened along the SLOPE path for a data set with 200 observations and 5000 features.

The Hessian Screening Rule

In this paper (Larsson and Wallin 2022) we continued our work on screening rules, but for the lasso instead.

Our contribution: a new rule that uses second-order information to better predict the support along the regularization path.

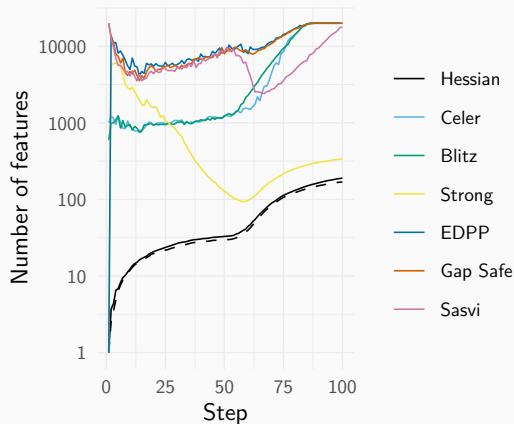


Figure 7: Number of features (predictors) screened along the SLOPE path for designs with varying correlation (ρ).

Benchopt (Moreau et al. 2022) strives to make benchmarking easy, transparent, and reproducible.

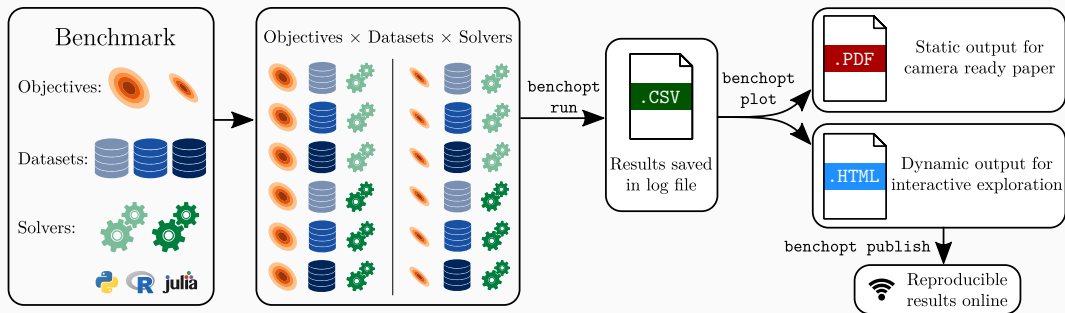


Figure 8: How Benchopt works.

Ongoing Work

Regularization And Normalization

- Normalization is essential for regularized methods, but there is almost no work on the topic.
- What effects do different types of normalization have on the solutions of regularized methods?

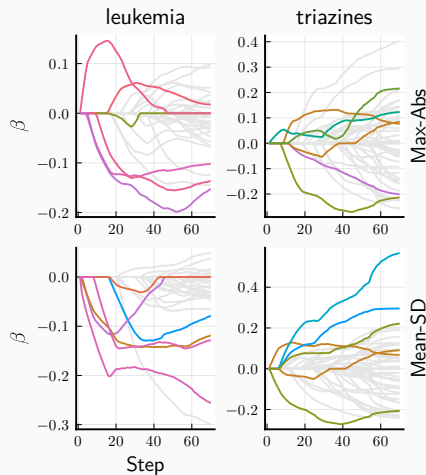


Figure 9: Lasso paths for two types of normalization.

Thank you!

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Coordinate Descent Steps

When updating the k th cluster, we let

$$\beta_i(z) = \begin{cases} \text{sign}(\beta_i)z, & \text{if } i \in \mathcal{C}_k, \\ \beta_i, & \text{otherwise.} \end{cases}$$

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Minimizing the objective in this direction amounts to solving the following one-dimensional problem:

$$\min_{z \in \mathbb{R}} \left(G(z) = P(\beta(z)) = \frac{1}{2} \|y - X\beta(z)\|^2 + H(z) \right),$$

where

$$H(z) = |z| \sum_{j \in \mathcal{C}_k} \lambda_{(j)_z^-} + \sum_{j \notin \mathcal{C}_k} |\beta_j| \lambda_{(j)_z^-}$$

is the *partial sorted ℓ_1 norm* with respect to the k -th cluster and where we write $\lambda_{(j)_z^-}$ to indicate that the inverse sorting permutation $(j)_z^-$ is defined with respect to $\beta(z)$.

The Partial Sorted ℓ_1 Norm

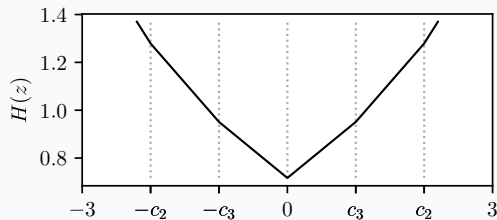


Figure 10: The partial sorted ℓ_1 norm with $\beta = [-3, 1, 3, 2]^T$, $k = 1$, and so $c_1, c_2, c_3 = (3, 2, 1)$.

How Do We Minimize Over One Cluster?

The optimality condition, using the directional derivative, is

$$\forall \delta \in \{-1, 1\}, \quad G'(z; \delta) \geq 0,$$

with

$$\begin{aligned} G'(z; \delta) &= \delta \sum_{j \in \mathcal{C}_k} X_{:,j}^\top (X\beta(z) - y) \\ &\quad + H'(z; \delta). \end{aligned}$$

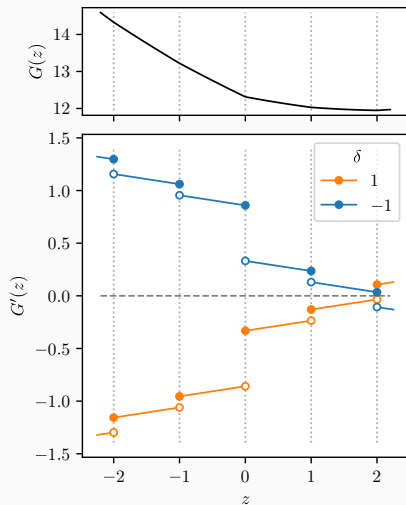


Figure 11: G and its directional derivative $G'(\cdot; \delta)$.

The SLOPE Thresholding Operator

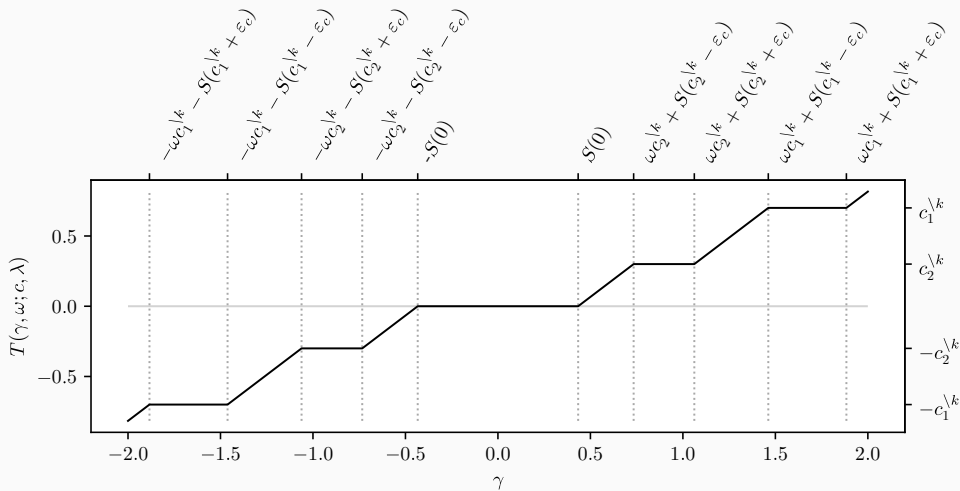


Figure 12: The SLOPE Thresholding Operator